

EDUCATION

Massachusetts Institute of Technology

Expected May 2027

Bachelor of Science in Computer Science and Electrical Engineering, Minor in Economics

Cambridge, MA

Relevant Coursework: Data Structures and Algorithms, Machine Learning, Probability, Linear Algebra, Computation Structures, C and Low-level Programming

Affiliations: MIT Entrepreneurship Club, MIT NSBE, Colorstack, MIT NEET, MLT

TECHNICAL SKILLS

Languages: Python, Go, C/C++, TypeScript, JavaScript, Java, SQL, R, MATLAB, RISC-V Assembly

Frameworks/Tools: FastAPI, Docker, Kubernetes, Google Cloud Platform, AWS, Git, WebSockets

Libraries: NumPy, pandas, TensorFlow, PyTorch, Plotly, LangChain, scikit-learn, Matplotlib, Seaborn

Databases: PostgreSQL, Redis, MongoDB, Firebase

EXPERIENCE

Expedia Group

May 2026 - Present

Machine Learning Engineering Intern

Austin, TX

- Built **Cross-EG Bench**, an internal embedding-model evaluation framework used across Expedia Group's lines of business, enabling reproducible retrieval benchmarking on LLM-judged and human-annotated datasets using Recall@K, MRR@K, and NDCG@K.
- Designed an end-to-end pipeline transforming raw labeling CSVs into versioned **Parquet** datasets and executing embedding and **FAISS** retrieval experiments on **Databricks GPU clusters**, with artifacts managed in **Unity Catalog**.
- Developed a **React 19** and **FastAPI** interface for comparing multiple experiments side by side, deployed internally on Expedia Group's RCP platform.

MIT Media Lab / Latent Lab

Jun 2026 - Present

Undergraduate Researcher, AI File Systems

Cambridge, MA

- Researching **AI-native file-system** concepts for Filescape, prototyping retrieval workflows that combine **embeddings**, **metadata extraction**, and **semantic search**.
- Built an **LLM-judge** pipeline to standardize and score free-text lookup responses, computing **precision/recall/F1** against a large topic taxonomy.

Massachusetts Executive Office of Technology Services and Security (EOTSS)

Jun 2025 - Aug 2025

Software Engineering Intern

Chelsea, MA

- Designed and deployed a hosting recommendation platform used by 20+ Massachusetts agencies, integrating intelligent assessment logic and cloud-readiness scoring.
- Built an automated submission-tracking system with Flask-Mail, reducing status-related support requests by 40% and improving submission visibility.

Jane Street IN FOCUS Fellowship

Jan 2025

Software Engineering Fellow

New York City, NY

- Engineered a robust trading program in Python to manage market positions and update fair prices in response to real-time market data.

MIT Institute for Data, Systems and Society

Sep 2024 - Present

Undergraduate Researcher

Cambridge, MA

- Utilized **Python** and web scraping tools (e.g., **BeautifulSoup**, **Selenium**) to extract over 600,000 JSON records from international museum websites, creating a detailed dataset of Ottoman Empire clothing artifacts.

PROJECTS

TrueSplit | Go, PostgreSQL, Redis, Next.js, SSE | truesplit-wheat.vercel.app

Jun 2026 - Present

- Built a group expense-splitting engine on a double-entry ledger with idempotent **Go** APIs, using Redis and PostgreSQL to ensure exactly-once effects under concurrent duplicate requests.
- Enforced zero-sum and group-solvency invariants at the database level via Postgres constraints rather than application logic alone, and shipped a **Next.js** dashboard with live balance updates over Server-Sent Events, deployed on Render and Vercel.

SentinelStream | Go, Redis Streams, PostgreSQL, React, deck.gl/WebGL | sentinelstream-5qls.onrender.com

Jul 2026

- Built a real-time fraud detection pipeline in **Go** using Redis Streams consumer groups and five explainable rule-based signals, including GeoHash-based impossible-travel detection; found and fixed two distributed-systems reliability bugs through live production monitoring, restoring correct backpressure behavior.